

Bounded-Excess Cotype for Gaussian Random Polytopes

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Status

Status: partial result, likely valid. This solves the bounded vertex-excess subcase $N - n \leq M$ of the arbitrary- N cotype problem in arXiv:2603.04749, with constants depending on M .

Source Problem

Huang and Tikhomirov prove in [1] that, when $K' \geq N/n \geq K > 1$, the Gaussian polytope

$$P_{N,n} = \text{conv}\{\pm X_i : 1 \leq i \leq N\} \subset \mathbb{R}^n$$

generates, with high probability, a normed space $(\mathbb{R}^n, \|\cdot\|_{P_{N,n}})$ of finite dimension-independent cotype. Their Open Problems section asks whether there are universal $q, C_q < \infty$ such that the same conclusion holds for every $N \geq n$. The same section asks the analogous question for random orthogonal projections of the standard cross-polytope.

1.4. Open problems. The assumptions on the number of generating vectors N used in this paper, exclude the asymptotic settings $\frac{N}{n} \rightarrow 1$ and $\frac{N}{n} \rightarrow \infty$. In particular, the condition $\frac{N}{n} = 1 + \Omega(1)$ in our argument is employed to obtain in-radius bounds which, in turn, provide a one-sided control of relative magnitudes of $\|\cdot\|_{P_{N,n}}$ and the standard Euclidean norm. It would be of interest to remove the constraints on relative magnitude of N and n in the cotype estimate. Further, one can ask about the optimal cotype constant in our model of randomness, as well as cotype estimates for random projections of a high-dimensional cross-polytope.

Problem (Gaussian convex hulls with arbitrary number of vertices). *Do there exist $q, C_q < \infty$ such that for every $N \geq n$, with high probability the space $(\mathbb{R}^n, \|\cdot\|_{P_{N,n}})$ has cotype q with constant at most C_q ?*

Problem (Optimal cotype of Gaussian convex hulls). *Is it true that for every constant $\varepsilon > 0$ there is $C_\varepsilon < \infty$ such that with high probability the space $(\mathbb{R}^n, \|\cdot\|_{P_{N,n}})$ has cotype $2 + \varepsilon$ with constant at most C_ε ?*

Problem (Random projections of cross-polytopes). *Let $N \geq n$, and denote by $\tilde{P}_{N,n}$ the orthogonal projection of the standard cross-polytope in $\mathbb{R}^N$ onto a uniform random n -dimensional subspace. Is it true that $\tilde{P}_{N,n}$ has finite (dimension-independent) cotype with probability $1 - o(1)$?*

Partial Result

Theorem 1 (bounded excess). *Let $M \geq 0$ be fixed. Let $N \geq n$, let $X_1, \dots, X_N$ span $\mathbb{R}^n$, and assume $N - n \leq M$. Then $(\mathbb{R}^n, \|\cdot\|_{P_{N,n}})$ has cotype 2 with a constant depending only on M . In particular, for Gaussian X_i this holds almost surely.*

Corollary 1 (bounded-codimension projected cross-polytopes). *Let $M \geq 0$ be fixed. If $L \subset \mathbb{R}^N$ is any n -dimensional subspace with $N - n \leq M$, then the orthogonal projection of the standard cross-polytope B_1^N onto L defines on L a normed space of cotype 2 with constant depending only on M .*

Idea of the Proof

The missing near-square case is not random at bounded excess. The Gaussian polytope is the image of the ℓ_1^N unit ball under the map $Ae_i = X_i$. Thus the associated normed space is the quotient $\ell_1^N / \ker A$. When $N - n$ is bounded, this quotient is by a bounded-dimensional subspace. Auerbach's lemma gives a projection onto that kernel whose norm is controlled by its dimension, so the quotient is uniformly isomorphic to a subspace of ℓ_1^N . The cotype 2 of ℓ_1 then transfers to the quotient.

Proof

We use two standard facts. First, ℓ_1 has cotype 2; denote one admissible cotype constant by C_{ℓ_1} . Second, if Y is D -isomorphic to a subspace of a space of cotype q with constant C_q , then Y has cotype q with constant at most DC_q .

Lemma 1 (finite-dimensional quotients). *Let X be a Banach space of cotype q with constant C_q , and let $F \subset X$ be an m -dimensional subspace. Then X/F has cotype q with constant at most $(m+1)C_q$.*

Proof. If $m = 0$ there is nothing to prove. For $m \geq 1$, choose an Auerbach basis $(u_j, u_j^*)_{j=1}^m$ of F , so $\|u_j\| = \|u_j^*\| = 1$ and $u_i^*(u_j) = \delta_{ij}$. Extend each u_j^* to a norm-one functional $\tilde{u}_j^* \in X^*$ by Hahn–Banach, and define

$$Px = \sum_{j=1}^m \tilde{u}_j^*(x) u_j.$$

Then P is a projection from X onto F and $\|P\| \leq m$. The map

$$T : X/F \rightarrow \ker P, \quad T(x + F) = (I - P)x,$$

is well-defined, linear, and bijective. Moreover $\|T\| \leq \|I - P\| \leq m + 1$, while the inverse map $\ker P \rightarrow X/F$ has norm at most 1, because the quotient norm never exceeds the norm of a representative. Thus X/F is $(m+1)$ -isomorphic to the subspace $\ker P$ of X . Cotype passes to subspaces and then across this isomorphism with the stated distortion loss. $\square$

Now set $A : \ell_1^N \rightarrow \mathbb{R}^n$ by $Ae_i = X_i$. Since the X_i span $\mathbb{R}^n$, the map A is onto and

$$A(B_1^N) = \text{conv}\{\pm X_i : 1 \leq i \leq N\} = P_{N,n}.$$

Therefore, for $x \in \mathbb{R}^n$,

$$\|x\|_{P_{N,n}} = \inf\{\|a\|_1 : a \in \mathbb{R}^N, Aa = x\},$$

which is exactly the quotient norm on $\ell_1^N / \ker A$. Since

$$\dim \ker A = N - n \leq M,$$

the lemma applied to $X = \ell_1^N$ gives cotype 2 with constant at most $(M + 1)C_{\ell_1}$. For independent standard Gaussian vectors, spanning $\mathbb{R}^n$ holds almost surely when $N \geq n$.

The projected cross-polytope statement is identical: an orthogonal projection $Q : \ell_1^N \rightarrow L$ maps B_1^N onto the projected cross-polytope, and the associated normed space is $\ell_1^N / \ker Q$ with $\dim \ker Q = N - n \leq M$.

Limitations

The constants depend on the excess $M = N - n$. Hence this does not settle the full arbitrary- N problem, which asks for constants uniform over all $N \geq n$. It also does not cover regimes where $N - n \rightarrow \infty$, including the large-aspect random-projection problem isolated in the source paper and in the previous attempt note for this target.

Novelty and Duplicate Check

The run indexes contained one previous attempt for arXiv:2603.04749 and no solution packet for this bounded-excess subcase. Web searches on 2026-06-29 for the exact arXiv id and title, for bounded-codimension/random polytope cotype phrases, and for finite-dimensional quotient/cotype variants found the arXiv source page and standard Auerbach projection facts, but no direct later solution of this subcase. The proof above is deterministic and uses only standard Banach-space facts.

Human Review Recommendation

Review the finite-dimensional quotient lemma and the identification of $P_{N,n}$ as a quotient image of B_1^N . If those are accepted, the bounded excess partial result is complete.

References

- [1] H. Huang and K. Tikhomirov, *Cotype of random polytopes*, arXiv:2603.04749, 2026.
- [2] B. Maurey and G. Pisier, *Séries de variables aléatoires vectorielles indépendantes et propriétés géométriques des espaces de Banach*, Studia Math. 58 (1976), 45–90.
- [3] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I and II*, Springer, 1996.